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The Pavitt Taxonomy, revisited: patterns of innovation in manufacturing and services

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Abstract In this article we discuss how to summarize the persistent and large heterogeneity in innovative behaviour and economic performance. A revision of the Pavitt (1984) Taxonomy—covering manufacturing and services, as well as ICT activities—is proposed as a key tool for identifying common characteristics and diversities in patterns. An extensive analysis of innovation survey data, on sources, objectives, inputs and outcomes of innovation, allows us to test alternative industry groupings, leading to an extensive assessment of a Revised Pavitt Taxonomy that is able to capture major structural differences in the relationship between innovation and performance. As industries' classifications has changed from NACE Rev. 1 to NACE Rev. 2 in 2008, a revised Pavitt Taxonomy is also provided for the new industries.

Keywords Innovation · Innovation surveys · Pavitt Taxonomy

JEL Classification O30 · O33

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1 Introduction

Studies[1] on firms and industries face the challenge of understanding the sources, evolution and persistence of the very high heterogeneity found in key characteristics. The standard approach of mainstream economics—based on a "well-behaving" representative firm operating in well-defined market contexts—can hardly be reconciled with the extent of such heterogeneity.[2] Evidence from Bottazzi et al. (2010) for the period 1991–2004 shows that (labour) productivity differentials within French and Italian manufacturing firms in the same three digit sector, measured as ratios between the 95th percentile and the 5th percentile, range from 2 to 9. In US manufacturing, considering firms in the same 4 digits industries, Syverson (2011) reports an average p90-p10 difference in (log total factor) productivity around 2. This heterogeneity appears to be pervasive across countries and persistent over time; similar results have emerged from studies on firm size, efficiency, profitability, innovation, and growth rate (Moreno and Coad 2015; Colombelli et al. 2014a).

The search for alternatives to the mainstream approach needs to identify the key factors shaping heterogeneity while getting rid of the "noise" of individual observations, so that causal relationship can be explored.

The Pavitt Taxonomy (Pavitt 1984) considered the nature, sources and patterns innovation, as well as firm size and market structure, identifying four classes: Science Based Industries, Specialized Suppliers Industries, Scale Intensive industries, Supplier Dominated industries. Its empirical base was the Science Policy Research Unit (SPRU) database on innovation in UK firms, and it convincingly identified regularities and differences among firms and industries, becoming a milestone in innovation and business studies. It was developed in parallel to work on evolutionary/Neo-Schumpeterian approaches (e.g. Freeman 1982; Dosi 1982, 1988) exploring the importance of diversity in economic activities. The original work was focused on manufacturing (putting the entire service economy in the suppliers dominated class); a more recent literature has used a similar approach for investigating innovative activities in services (Tidd et al. 2005; Gallouj and Savona 2009). Efforts to extend or redefine industrial or firms taxonomies have usually focused on the production/innovation side, trying to cluster economic units according to their characteristics and strategies. However, one of the advantages of the Pavitt Taxonomy was its ability to fit the diversity of industries in the context of the relationship between innovation and economic performance.

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[2] Although analytically elegant, this approach is lacking empirical grounds; it assumes that all firms are forced to use the best available information, or face exit; empirically, however, this selection process appears to be ineffective (Bottazzi et al. 2010).

This article reassesses the Pavitt Taxonomy and extends it to service industries on the bases of systematic empirical evidence; using a complex array of techniques, we provide a grouping that incorporates varieties of input and innovative strategies and differences in the nexus innovation-performance. Focusing on regularities at the industry level for both manufacturing and services, we propose and test a Revised Pavitt Taxonomy. A major novelty of this article is the systematic and detailed use of empirical evidence and statistical tests, using the rich information provided by European Community Innovation Surveys (CIS).[3] They account for different dimensions of innovative efforts, addressing the nature of innovation, its sources, strategies, cost, outcomes and economic relevance. This analysis is based on the sectoral innovation database (SID), developed at the University of Urbino, which provides data from three waves of the CIS for 38 manufacturing and services industries in eight European countries (Pianta et al. 2015). Our methodology uses a variety of statistical tools—descriptive statistics, distributional tests, principal component and regression analyses—to test the stability of the Revised Pavitt Taxonomy over time and across countries.

Our findings confirm that the four classes originally identified by Pavitt still hold after a proper placement of service industries—including ICT intensive activities. Moreover, the Revised Pavitt Taxonomy is able to account for the diversity in the determinants of innovation and in the innovation-performance link across European industries.

This article proceeds as follows: Sect. 2 discusses heterogeneity and introduces our approach; Sect. 3 reviews the literature on applications and extensions of the Pavitt Taxonomy; Sect. 4 presents the Revised Pavitt Taxonomy and its empirical grounding, Sect. 5 provides some econometric tests. Section 6 briefly concludes.

2 Dealing with heterogeneity: why industries matter for understanding innovation

Research on firms and industries has long faced the challenge of summarizing in meaningful ways the key characteristics of innovation and economic processes, considering commonalities and differences in observed patterns for relevant units of analysis. We identify four broad approaches.

First, neoclassical economics has assumed the existence of a "representative firm" whose behaviour—as a profit-maximising agent operating in a well-defined environment, with freely available information—can be generalised to all firms and industries, neglecting the existence of structural differences.

[3] CIS are surveys representative of the universe of firms in a country. They are based on the Oslo Manual (OECD 2005). Surveys have been carried out in a fully comparable way through the European Union. The second, third and fourth waves used in the present work were carried out every 4 years (1996, 2000, 2004), each one covering also the two preceding years. For a discussion, see Eurostat (2008). Innovation surveys allow researchers to overcome many of the limitations of traditional innovative indicators, such as patents (Archibugi and Pianta 1996a, b), normally too biased towards science based sectors, but are subject to the critique of being 'subjective' surveys (Smith 2005). An assessment of innovation surveys carried out in emerging countries is in Bogliacino et al. (2012).

Second, studies with an evolutionary perspective have emphasised the heterogeneity of the universe of firms, showing major structural differences in terms of internal characteristics—competences, capital, labour, etc.—and economic performances. These studies have also shown that observable characteristics—such as size—are poor predictors of corporate performance due to structural differences across firms, "fat tail" distributions, and a sizable amount of uninformative noise (Dosi 2007; Bottazzi et al. 2011; Dosi et al. 2012).

Third, industry studies have taken an intermediate perspective, emphasising sectoral specificities in manufacturing and services in the characteristics of firms and technologies and exploring sources, dynamics and performances at such level. While intra-sector heterogeneity may be sizable, industries share the same technological opportunities, nature of knowledge, appropriability conditions and market structure. While firm level studies are usually based on panels including a relatively small number of firms that are not representative of the universe, industry level studies account for the totality of business in a given sector, leveling off gains and losses that may occur at the firm level, considering the changes in the structure of the economy and the role of demand (Breschi et al. 2000; Malerba 2004; Bogliacino and Pianta 2011, 2013; Bogliacino et al. 2013). In this literature, a special mention should be done with respect to the possibility of characterizing the knowledge base of companies and sectors and use it as a predictor of performance (Colombelli and Quatraro 2014; Krafft et al. 2014a, b; Colombelli et al. 2014b).

Fourth, the influential taxonomy proposed by Pavitt (1984)—that can be referred to either firms or industries—has opened up a different perspective, based on the nature and sources of technological change, of production systems, and market structures. This approach identifies, in particular, the differences in the ways technology is developed and used.

We build on the latter approach and apply it at the industry level, in order to summarise in a meaningful way the heterogeneity of firms' behaviour. Indeed, most applications of the Pavitt Taxonomy have been carried out on industries. After discussing the possibility of jointly considering the characteristics of manufacturing and services, we proceed to the empirical investigation, where we map the distribution of variables on the *intensity* and *diffusion* of innovation and performance at the industry level using innovation survey data.[4] We then explore commonalities in the distribution of industries with regard to their innovative activities, testing alternative groupings and obtaining a Revised Pavitt Taxonomy extended to service industries.

In the analysis of innovation data, the use of a double mapping, from firms to industries and from the latter to Pavitt classes, is a key step in our strategy; the methodological reasons behind this approach are the following:

[4] Innovation survey variables are either expressed as average intensity of particular activities (e.g. average R&D expenditure per employee within firms of a sector) or are measured as share of firms with a particular effort (e.g. share of firms within an industry carrying out product innovation) highlighting the diffusion of particular activities.

(a) Averaging at the industry level corrects for the bias associated with the subjective nature of questions. The direction of the error is non-systematic across firms in the same industry, filtering out an important source of noise;
(b) As stressed by Peneder (2010), reporting information on the share of firms, which carry out a certain activity, together with the sectoral intensity, is a meaningful way to summarise the underlying distribution;
(c) The industry level allows capturing some of the characteristics that operate beyond the scale of individual firms. They include the role of *demand pull*; at the company level demand is perfectly elastic due to the possibility of "stealing" market shares from other firms, while at the industry level it is a binding constraint. Firms are affected by the increasing vertical disintegration of production and occupy different positions in the value chain, exacerbating heterogeneity, while an industry level approach allows a more integrated perspective on industrial change.[5]

3 Rethinking the Pavitt Taxonomy

3.1 Pavitt's definition and the literature

Building on information on UK firms collected at SPRU, Pavitt (1984) identified the following four groups:

1. *Science Based industries* (*SB*) they include sectors where innovation is based on advances in science and R&D (such as the pharmaceutical and electronic industries), where research laboratories are important, leading to intense product innovation and a high propensity to patent.
2. *Specialized Suppliers industries* (*SS*) they include the sectors producing machinery and equipment: their products are new processes for other industries. R&D is present but an important innovative input comes from tacit knowledge and design skills embodied in the labour force. Average firm size is small and innovation is carried out in close relation with customers.
3. *Scale Intensive industries* (*SI*) they include industries were scale economies are relevant (automotive and basic metals) and a certain rigidity of production processes exists, so that technological change is usually incremental. Important process innovation coexists with new product development.
4. *Supplier Dominated industries* (*SD*) they include traditional sectors (such as food and textile) where small firms are prevalent and technological change is introduced through the inputs and machinery provided by suppliers from other industries. Firms in this group do not carry out much R&D or other innovative activities.

[5] One of the critiques to the use of industry level data is that firms with similar technologies may potentially end up in different industries because of product characteristics (Archibugi 2001). However, this potential error (rather limited) is likely to be corrected by the subsequent mapping of sectors into Pavitt groupings. In fact, such sectors are likely to end up in the same Pavitt class.

A large literature has applied this taxonomy in order to investigate the sources of heterogeneity in innovation and performance. A major finding is that Pavitt classes indeed fit well the diversity in the relationship between innovation and economic outcomes, using different measures and levels of analysis. Dosi et al. (2008) applied Pavitt classes to Italian microdata to study the size distribution and the effects of technological change. Various contributions used Pavitt groupings in investigations of trade specialisation, appropriability and regional patterns of innovation.[6] The Pavitt Taxonomy has been combined with CIS data in order to explore various facets of innovative patterns.[7] At the industry level, patterns of innovation and market share dynamics are investigated by Laursen and Meliciani (2000) and the diversity in input–output relationships is studied by Di Cagno and Meliciani (2005). National patterns have also been assessed by several studies.[8] An earlier review can be found in Archibugi (2001).

This large literature has provided a strong empirical support across data sources and countries on the ability of the taxonomy to offer insights on key aspects of innovative and economic performance. When the focus of analysis is restricted to more specific issues, more refined industry classifications have been adopted.[9] Additional studies have developed alternative classifications based on different units of analysis (e.g. regions, clusters, goods, etc.).[10] The lesson emerging from such extensive review of taxonomic efforts is that the Pavitt Taxonomy is a robust conceptual tool, versatile in its possible applications, with wide data availability and low risk of aggregation bias. However, it has been criticized for being too focused on technology and for neglecting the heterogeneity of services (Gallouj 1999, 2002).

6 On trade see Dosi et al. (1990) and Padoan (1998). On property rights see Dosi et al. (2006). Maggioni et al. (2011) applied Pavitt classes to patenting activity of Italian NUTS3 regions in order to investigate knowledge transfers and spillovers.

7 Heidenreich (2009) explored innovation in low and medium technology industries (CIS wave four); Castellacci (2008, 2009) explored industry patterns and cross country variability in systems of innovation (CIS wave two). Resmini (2000) explored the determinants of FDIs in the European Union according to different Pavitt groupings; Freel (2003) studied the effects of collaboration on the innovativeness of SMEs located in Northern England and Scotland; Smith et al. (2002) addressed the R&D performance in a sample of Danish firms.

8 See Pavitt et al. (1989) for the United Kingdom; Marsili and Verspagen (2002) for the Netherlands using data for manufacturing from the Business Register and CIS; Dutrenit and Capdevielle (1993) for Mexico using industrial census data.

9 A detailed analysis of this large literature is beyond the scope of this article; notable examples include the following: Lall (2000) maps industries according to the technological content of export; Guerzoni (2010) according to the nature of the demand pull motive to innovate; Castellacci (2008) according to innovation and the position in the vertical chain; O'Mahony and Vecchi (2009) according to factor and skill intensities; Montobbio (2003) according to R&D intensity; Peneder (2002) according to intangibles and human capital intensity; Peneder (2008) according to cost of experimentation; Raymond et al. (2006) use a parametric approach based on an estimated model.

10 Examples of taxonomies using different units of analysis include the following: Bardolet et al. (2010) classify business types; Fudenberg and Tirole (1984) business strategies; Di Berardino and Mauro (2010) regions; Baskaran and Muchie (2009) national systems of innovation; Dana et al. (2009) entrepreneurs; Iammarino and McCann (2006) economic clusters; Gallouj (1999) goods; Jindra (2005) multinational corporations; De Cleyn and Braet (2009) spin-offs and spin-out; De Jong and Marsili (2006) innovative small businesses; Love (2003) companies receiving FDIs.

Nevertheless it should be remembered that Pavitt was working on the SPRU database, which was focused on ‘innovations’, i.e. artefacts, which may explain the limited attention to services.

3.2 Extensions to services and ICTs

Moving from Pavitt’s approach, a large amount of work has recently been carried out on the patterns of technological change in services, which represent the dominant part of advanced economies. In a comprehensive survey, Gallouj and Savona (2009) summarize the state of the art and distinguish between studies that “assimilate” services to manufacturing and approaches emphasising differences. The latter were influenced by the conventional wisdom that services were less innovative (partly due to a measurement error, as documented by Gallouj and Savona 2009). Available evidence shows a substantial convergence of innovative activities between manufacturing and services; this can be the result of the evolution of industries’ life cycles, of manufacturing activities incorporating increasing service content, and of services being increasingly organised along large-scale industrial models. Moreover, the shift from Fordist to post-Fordist models of organisation of production has turned integrated hierarchical (manufacturing) companies into network-based corporations, outsourcing both manufacturing and service activities in more complex systems of organisation of production.

Important work on the extension of the Pavitt Taxonomy to services is found in Barras (1986), Gallouj (1999), Evangelista (2000), Miozzo and Soete (2001), Evangelista and Savona (2003) and Camacho and Rodriguez (2008); a systematic discussion is in Gallouj and Djellal (2010). While focusing on different aspects of service production and proposing slightly different allocations of services industries to the various classes, this literature broadly confirms the interest of a taxonomy based on the Pavitt approach.

Pavitt himself in a joint work with other two scholars has extended his classification to services proposing an additional group for Information Intensive industries (Tidd et al. 2005). The main technological feature of this class is the focus on the design, use and improvement of large ICT systems to process information.

Our approach builds on these findings, acknowledging the increasing convergence between manufacturing and services and testing the statistical robustness of the alternative groupings proposed; in particular we systematically compare the strengths of the classification based on the four original Pavitt classes and of its extension to five groupings. Our results suggest that the latter is less robust, as shown in Sects. 4 and 5; Information Intensive sectors behave in a manner, which is not statistically different from Scale Intensive ones. We therefore propose to combine them in a new Scale and Information Intensive (SII) class.

3.3 The Revised Pavitt Taxonomy

Our Revision of Pavitt Classes is based on the two digits Nace classification (Rev. 1), of manufacturing and services and is presented in Table 1. In 2008 the Revision 2 of the NACE classification started to be adopted. As no substantial series of data

Table 1 The Revised Pavitt taxonomy for manufacturing and services

Revised Pavitt Taxonomy	NACE Rev. 1
Science based	
Chemicals	24
Office machinery	30
Manufacture of radio, television and communication equipment and apparatus	32
Manufacture of medical, precision and optical instruments, watches and clocks	33
Communications	64
Computer and related activities	72
Research and development	73
Specialised suppliers	
Mechanical engineering	29
Manufacture of electrical machinery and apparatus n.e.c.	31
Manufacture of other transport equipment	35
Real estate activities	70
Renting of machinery and equipment	71
Other business activities	74
Scale and information intensive	
Pulp, paper and paper products	21
Printing and publishing	22
Mineral oil refining, coke and nuclear fuel	23
Rubber and plastics	25
Non-metallic mineral products	26
Basic metals	27
Motor vehicles	34
Financial intermediation, except insurance and pension funding	65
Insurance and pension funding, except compulsory social security	66
Activities auxiliary to financial intermediation	67
Suppliers dominated	
Food, drink and tobacco	15–16
Textiles	17
Clothing	18
Leather and footwear	19
Wood and products of wood and cork	20
Fabricated metal products	28
Furniture, miscellaneous manufacturing; recycling	36–37
Sale, maintenance and repair of motor vehicles and motorcycles; retail sale of automotive fuel	50
Wholesale trade and commission trade, except of motor vehicles and motorcycles	51
Retail trade, except of motor vehicles and motorcycles; repair of personal and household goods	52
Hotels and catering	55
Inland transport	60
Water transport	61

Table 1 continued

Revised Pavitt Taxonomy	NACE Rev. 1
Air transport	62
Supporting and auxiliary transport activities; activities of travel agencies	63

Two digit Nace classification, Rev. 1

with such grouping is so far available, the empirical tests have focused on the previous classification. However, building on original work for matching the two classifications (Perani and Cirillo 2015; Pianta et al. 2015), a Revised Pavitt taxonomy also for the Nace classification (Rev. 2) is presented in the Appendix (Table 8). The Revised Pavitt taxonomy has already been used in detailed investigations of the diversity in the relationships between innovation and employment (Bogliacino and Pianta 2010) and innovation and productivity (Bogliacino and Pianta 2011).

The assignment of manufacturing industries to each grouping follows the original Pavitt Taxonomy; the allocation of services is based on conceptual and empirical grounds. Communications, Research and Development and Computers leave little doubt on the closeness of such industries to the Science Based group. Secondly, real estate, renting of machinery and other business activities tend to include specific activities that are tailor made for clients, sharing the characteristics of Specialized Suppliers. Third, financial and insurance services, and activities auxiliary to them, are characterised by large firm size and an extensive adoption of ICT based machinery and are therefore included in the new Scale and Information Intensive class. Finally, wholesale and retail trade, and all transport services include very little internal innovative efforts and mainly rely on innovations provided by suppliers. The empirical support for this classification is provided in the next Section.

4 Empirical tests on the Revised Pavitt Taxonomy

4.1 Data

The data source for this work is the Sectoral Innovation Database (SID) developed at the University of Urbino. It includes most variables of three comparable waves of the CIS (CIS 2, 3 and 4), matched with a large array of information coming from other sources and available at the same sectoral level. The country coverage of the database includes seven European countries and one EFTA country—Germany, France, Italy, the Netherlands, Portugal, Spain, the United Kingdom and Norway. Data are available for the two-digit NACE classification (Rev. 1) of both manufacturing and service industries. The full description of the sources and methodology followed for the construction of the database is provided in the SID

Table 2 Sectoral innovation database: the variables

Variable description	Unit	Source
Share of firms introducing new products	%	CIS
Share of turnover from new or improved products	%	CIS
Share of firms innovating with the aim to open new markets	%	CIS
Share of firms applying for a patent	%	CIS
Share of firms defining clients as source of inn.	%	CIS
Share of firms performing in house R&D	%	CIS
In house R&D expenditure per employee	Thousands euros/empl	CIS
Total R&D expenditure per employee	Thousands euros/empl	CIS
Share of firms introducing new processes	%	CIS
Share of firms innovating with the aim to reduce labour cost	%	CIS
Share of firms defining suppliers of equipment as source of inn.	%	CIS
Share of firms introducing innovative machinery and equipment	%	CIS
Expenditure on innovative machinery and equipment per employee	Thousands euros/empl	CIS
Share of innovative firms	%	CIS
Total innovative expenditure	Thousands euros/empl	CIS
Compound rate of growth of value added	Annual rate of growth	STAN

Methodological Notes (Pianta et al. 2015).[11] Data for CIS 2, 3 and 4 are fully comparable and account for the total population of firms.

Innovation variables are either expressed as shares of firms—measuring the diffusion of a certain activity in the underlying distribution—or as thousand euros per employee—measuring the intensity of innovation expenditures (Table 2).[12]

4.2 Direct approach

In this Section we show how the taxonomy itself emerges through a data-driven, indirect approach. We try to disentangle the latent dimensions of innovative activity through the use of principal component analysis and we calculate the values for

[11] CIS data are collected by national statistical institutes in cooperation with Eurostat. The questionnaire is common to all countries and is based on the Oslo Manual (OECD 2005); the same holds for the methodology, which includes stratification of the sample by industry. Firms' responses have been reported to the two digits industry level with the use of proper weights, derived from the sampling procedure. The construction of the SID has been carried out by the University of Urbino through cooperation agreements with national data providers with access to their country's industry-level data—either national statistical institutes or research groups with authorisation to exchange data—in the case of CIS 2 and 3; CIS 4 data are available from Eurostat, except for the UK, whose data have been obtained from the national data provider. The assembling of the database has been carried out using a common protocol following the procedure in use at the National Statistical Offices. The latest CIS surveys—using the NACE (Rev.2) classification—have recently been included in the database using the conversion methodology developed in Perani and Cirillo (2015).

[12] When monetary values are used (as in R&D expenditure) nominal variables have been deflated using the GDP deflator from Eurostat (base year 2002) and a PPP conversion for non-euro countries (from Stapel et al. 2004).

Table 3 The eigenvectors (scores) associated with the retained components

Variable	First component (cost competitiveness)	Second component (technological competitiveness)
Share of innovative turnover	0.36	0.44
Share of firms introducing new machinery	0.45	−0.06
Share of in house R&D in total innovation expenditure	0.11	0.78
Share of firms innovating to reduce labour cost	0.47	−0.24
Share of firms indicating suppliers as source of innovation	0.42	−0.33
Share of firms introducing process innovation	0.49	0.07

these latent variables using as weights the estimated eigenvectors associated to the retained components.

We use variables in the same scale, namely shares. We include six variables: share of R&D in innovative expenditure (R&D plus new machinery) which captures the importance of innovative effort (this variable is higher for technology "generators" and lower for technology "adopters"); share of innovative turnover and share of process innovators to proxy innovative output; share of firms investing in new machinery for the structural dimension; share of firms indicating suppliers as source of innovation to capture the importance of adoption; finally, the share of firms that invest to reduce labour cost, to proxy for labour saving determinants.

According to Kaiser criterion—namely eigenvalues greater than one—we retain two components, with associated eigenvalues of 3.06 and 1.22, explaining cumulatively 71.36 % of the variance. The associated screenplot supports our choice: after the second dimension there is a change of slope in the curve. The Kaiser–Meyer–Olkin measure is 0.66. In Table 3 we show the associated eigenvectors rotated according to the standard normalization (sum of squares equal to one) that we use to estimate the values for the latent variables.

As discussed in Pianta (2001) it is possible to build on the original Schumpeterian distinction between product and process innovation identifying, respectively, a strategy of technological competitiveness, focused on opening up new markets, and a strategy of cost competitiveness, based on labour cost reduction.[13] The two retained components closely resemble this distinction: the former is proxied by component two, which is associated with R&D and product innovation, while the latter is described by component one, strongly associated with new machinery, process innovation and labour saving motive.

[13] The strategies are not mutually exclusive, since they can coexist in the same firm or industry. However, they have different effects on productivity and employment; sectors tend to be dominated by either one or the other strategy; for empirical assessments see Bogliacino and Pianta (2010, 2011, 2013).

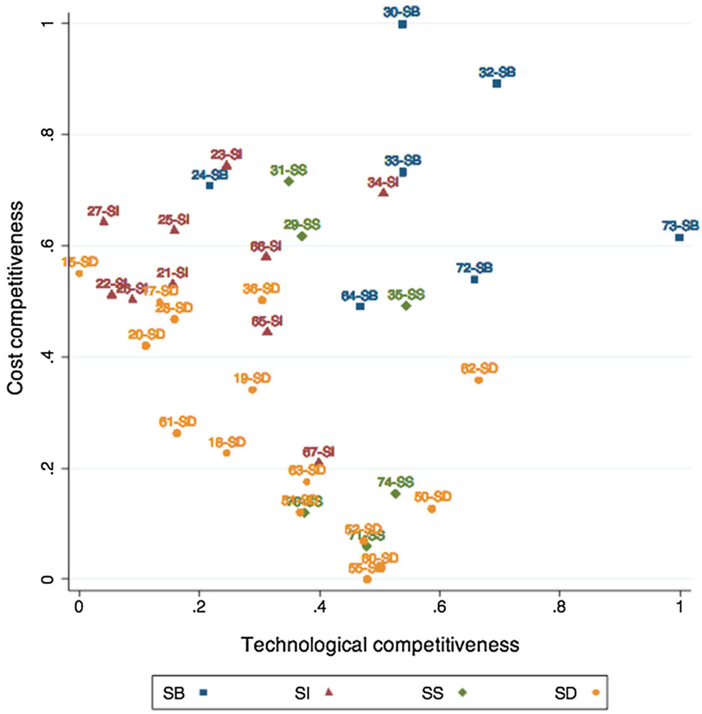

Fig. 1 Latent dimensions of innovative activity and Revised Pavitt Classes

We now use the coefficients shown in Table 3 to calculate the components, then we rescale on a zero–one basis and we plot the average value of the component calculated at industry level (averaging the eight countries). In order to appraise the robustness of the taxonomy, we plot the two components showing the NACE code of the observation together with the labels SB, SS, SI and SD to indicate the Revised Pavitt Taxonomy grouping to which each industry belongs. The results are shown in Fig. 1.

The picture is fairly consistent with the patterns discussed above. There is a clear hierarchy defined by the technological leadership of Science Based industries on the top right of the graph, and the low technological dynamism of Suppliers Dominated ones on the bottom left side. Scale and Information Intensive industries appear to rely more on cost competitiveness (top left cluster), while Specialised Suppliers occupy an intermediate position.

In order to further confirm the results, we statistically assess the robustness of this clustering by means of t-tests for each grouping: we test whether the means are significantly different for both components, using one class at a time against the rest of the dispersion. The test allows for unequal variances of the distributions. For SB the t statistics for the first and second component are respectively −21.29 and −14.89; for SI −10.49 and 15.00; for SD 19.89 and 5.04; finally for SS 4.14 and

−8.52. The equality of the means is always rejected against the alternative hypothesis of differences, with p value lower than 0.000 in all cases.

4.3 Indirect approach

As we discussed in Sect. 3.3 there are theoretical arguments supporting the allocation of different industries to the four groupings. The most delicate step in this analysis is the allocation of services to the four Revised Pavitt Classes. In Table 4 we provide the average value of some key variables. For completeness we also report the corresponding value of the same class restricted to manufacturing.

According to the data, the three service industries classified into SB have the largest R&D intensity (in thousand euros per worker): Communications 0.72; Computer 2.40; Research and Development 7.9. They also have the largest innovative share of turnover: respectively about 24, 26 and 35 %. These two variables document how important research is in the generation of new products (Pavitt 1984).

Specialised suppliers services have non-negligible R&D, considerable innovative turnover and are less focused on labour cost reduction, due to the role played by human resources and embodied knowledge into the innovative effort. This is coherent with the conceptualization of the corresponding manufacturing ones in Pavitt (1984), which stressed the importance of firms' specific skills with respect to processes of formalized search.

Financial intermediation, insurance and related financial industries (classified as SII) are characterized by scale economies and have some of the largest expenditure on machinery per employee: 0.76, 1.34, 2.35 thousand euros per worker; they also have the largest shares of firms with labour saving strategy, with a similar profile to manufacturing SII industries.

SD industries are clearly less intensive in overall innovative efforts; typically, they focus on process innovation and again they share the features of traditional producers in manufacturing.

In an effort to capture the specificity of industries at the core of the ICT revolution, Tidd et al. (2005) proposed to introduce a new class of Information Intensive sectors. Communications, financial sectors, retailers and publishers are presented as industries that organize production and work around new machineries able to process and diffuse information. In order to test the robustness of such a class we consider in this group the NACE (Rev. 1) classes of Communications, Financial Intermediation, Insurance and Pension Funding, Activities auxiliary to Financial Intermediation, Retail Trade and Printing and Publishing.

Tidd et al. (2005) emphasized the role of ICT equipment in shaping innovation processes in such a group. However, the same pattern may be found in Scale Intensive industries. On the other hand, Communication services appear to share the same strong R&D role as Science Based industries, while the low innovative intensity of Retail Trade is similar to the prevailing patterns of Suppliers Dominated industries.

Table 4 Innovative effort of services industries

Industry	R&D exp per employee	Machinery exp. per employee	Labour saving strategy (%)	Innovative turnover (%)
Science based—SB				
Average SB manufacturing	9.03	2.51	33.67	33.83
Communications	0.72	1.15	18.36	24.59
Computer and related activities	2.40	0.86	18.94	26.88
Research and development	7.9	1.18	22.83	35.56
Specialised suppliers—SS				
Average SS manufacturing	4.47	1.26	27.83	19.89
Real estate activities	0.17	0.4	15.01	5.53
Renting of machinery and equipment	0.13	0.68	14.90	8.31
Other business activities	0.22	0.12	15.04	11.01
Scale and information intensive—SII				
Average SII manufacturing	1.65	2.78	28.00	15.76
Financial intermediation, except insurance and pension funding	0.61	0.76	23.66	13.05
Insurance and pension funding, except compulsory social security	1.75	1.34	27.05	17.55
Activities auxiliary to financial intermediation	0.63	2.35	17.97	6.48
Suppliers dominated—SD				
Average SD manufacturing	0.50	1.11	24.09	12.84
Sale, maintenance and repair of motor vehicles and motorcycles; retail sale of automotive fuel	0.02	0.20	18.15	10.23
Wholesale trade and commission trade, except of motor vehicles and motorcycles	0.28	0.41	13.01	7.10
Retail trade, except of motor vehicles and motorcycles; repair of personal and household goods	0.10	0.13	10.85	4.44
Hotels and catering	0.02	0.17	13.06	5.67
Inland transport	0.11	0.66	11.47	6.44
Water transport	0.38	4.21	15.72	4.19
Air transport	0.07	2.23	16.38	13.49
Supporting and auxiliary transport activities; activities of travel agencies	0.65	1.28	15.70	7.06

Average values for European industries over the three CIS waves

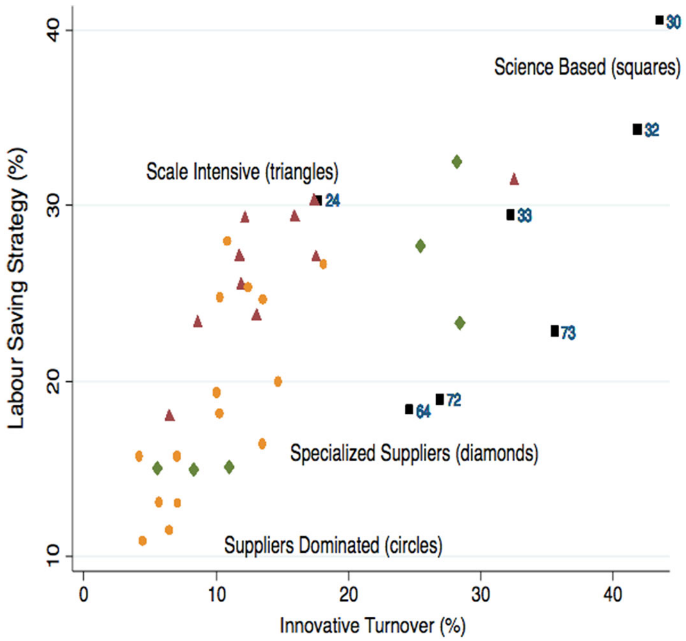

Fig. 2 Innovation strategies. European industries

The two alternatives we want to test are (a) the coherence of the Information Intensive Sector as a distinct grouping versus (b) the distribution of such services in the previous four Pavitt classes described above.

We use a standard t test to investigate whether there is a difference in means between the industries 21, 23, 25–27, 34 (manufacturing Scale Intensive) and sectors 22, 65–67 (Information Intensive ones). We look at the distribution of both the share of firms who introduce new machinery and the expenditure on new machinery. The equality is not rejected, at 5 % for the former variable ($t = -1.06$, p value 0.28) and at 1 % for the latter ($t = -2.44$, p value 0.02). These results provide strong evidence in favour of the robustness of the original four classes taxonomy; further tests are carried out in Sect. 5.

Further evidence comes from the identification of innovative strategies. We use the same definition of technological and cost competitiveness introduced above. The variables we use in order to operationalise such strategies are the share of innovative turnover (technological competitiveness) and the share of firms innovating to reduce labour cost (cost competitiveness). Results are shown in Fig. 1, where we plot the average values of industries across the eight countries. Revised Pavitt classes appear to be effective in isolating the underlying patterns: the heterogeneity within each class is lower than the variability between classes.

Figures 1 and 2—which are based on different sets of variables—show strong similarities suggesting the robustness of the identified patterns.

Table 5 A multinomial logit analysis for the Revised Pavitt taxonomy: odds ratios

	Odds ratio		
	SB	SII	SS
Share of firms introducing new products	0.083 (0.021)***	−0.018 (0.018)	0.030 (0.020)
Share of turnover from new products	0.081 (0.020)	0.002 (0.020)	0.060 (0.017)***
In-house R&D expenditure (thousand euros per empl)	1.099 (0.333)***	0.773 (0.321)**	1.044 (0.334)***
Share of firms introducing new process	−0.035 (0.024)	0.061 (0.017)***	−0.061 (0.024)**
Share of firms aiming to reduce labour cost	−0.068 (0.014)***	−0.017 (0.012)	−0.015 (0.011)
Machinery expenditure (thousand euros per empl)	0.138 (0.214)	0.533 (0.190)***	−0.249 (0.221)
Demand: rate of growth of value added	0.131 (0.0418)***	0.090 (0.045)**	0.113 (0.034)***
Constant	−4.883 (0.620)***	−2.517 (0.462)***	−1.599 (0.519)***
N obs	428	Pseudo R2	0.32

Dependent variable: Revised Pavitt classes. Odds expressed relatively to the Suppliers Dominated industries. Standard errors in parentheses

* Significant at the 90 % level; ** significant at 95 %; *** significant at 99 %

4.4 Testing differences across classes: a multinomial logit analysis

We now move to a third approach, where we start from our Revised Pavitt Taxonomy and we explore the contribution of key variables in allocating an observation to a certain class. Methodologically, we use (unordered) multinomial logit. In Table 5 we show the odds ratio using the omitted class of Suppliers Dominated as reference group.

Typically, according to Pavitt's conceptualization, we would expect significant R&D effort and product innovation to characterize sectors belonging to the SB class; we expect new machinery to be important in the SII due to the existence of scale economies; we expect the SS sectors to rely on both new products and new processes.

The results are coherent with the framework. By computing marginal effects, we can see how a marginal change in one of the variable affects the likelihood for an industry to be classified in each class (including the omitted category, SD). An increase in the share of product innovators by 1 % in a given industry increases the probability of entering the Science Based class by 0.8 %, while the same change in process innovators (labour cost savers) reduces it by 0.5 %. A 1 % increase in the share of turnover from new products raises the same probability by 0.6 %; a 1 % increase in the share of firms aiming to reduce labour cost reduces the probability by 1 %.

In the case of Scale Intensive industries, raising by 1000 euros the expenditure on machinery per employee increases the probability of being classified into this group by 15.1 %. For Specialised Suppliers innovative turnover is playing the main role (0.7 %)

together with the expenditure on machinery, which reduces by 9.9 % the probability of being classified here. Finally, 1000 more euros per employee spent in R&D lowers the probability of being classified among Suppliers Dominated by 13 %.

In the Appendix (Table 9), we report also the Relative Risk Ratio, which have a more direct interpretation with respect to the baseline outcome. In Table 10 we provide the results for services sectors only: while the statistical power drops significantly due to the number of observations, qualitatively the results do not change. The main difference is the effect of process innovation for SII class, and the role of demand.

5 Innovation and performance in Revised Pavitt Classes

5.1 Investigating innovative performances

In this Section, we focus on the relationship between innovative effort and performance; we test whether the performance of each class is shaped by a different set of determinants. As a dependent variable we use the share of innovative firms in the industry; as explanatory variables we use a series of indicators that capture the objectives, sources and inputs of innovation, that are considered to be at the root of diversity in innovative behaviour. In particular we consider the share of firms in the sector which innovate to reduce labour cost and to open up new markets (proxies for the objectives), the share of firms which perform R&D internally and the share of firms which introduce new machinery (proxies for technology generation and adoption) and total expenditure per employee as a measure of intensity. We include country dummies, testing their significance with a Wald test.

The econometric strategy is to run a regression that allows for heterogeneous coefficients across Pavitt classes and then to test the differences in the coefficients (through a Wald test).

Since data cover three time periods (the three CIS waves), we test for the presence of an error component structure. We first run a Random Effects Model in order to compute a Breusch and Pagan Lagrangian Multiplier test. The equality of the variances is not rejected (the chi^2 statistics is 1.14 and the p value is 0.28). Given the results of the diagnostic tests, we maintain OLS as baseline estimation.[14] The results are presented in Table 6.

The Wald test rejects the hypothesis of non-systematic differences among coefficients (the chi^2 statistics is 286.89, p value 0.00).

A number of important findings emerge. The search for new markets is highest in the Science Based group and is significant, with lower coefficients, for the

[14] Since the dependent variable is in percentage scale, we consider the possibility that the truncated nature of the range of variation could affect the estimate. We run a random effect Tobit and test for the presence of unequal variance for the random effect. The null hypothesis of equal variances is again not rejected (χ^2 statistics is 0.50, p value 0.23). The coefficients from the Tobit regression are equal to the OLS one (up to the second decimal), with slightly different standard errors (mainly due to bootstrapping to correct for heteroskedasticity in the former estimation), which do not affect the test of significance for any of the regressors. Indeed, the use of industry level data already correct for the problem of data censoring since there is no probability mass on the extreme values.

Table 6 The determinants of the share of innovative firms

	(1) SB	(2) SS	(3) SII	(4) SD
Share of firms performing in house R&D	0.410 (0.079)***	0.458 (0.054)***	0.265 (0.062)***	0.406 (0.072)***
Share of firms aiming to open new markets	0.509 (0.145)***	0.343 (0.088)***	0.328 (0.113)***	0.128 (0.090)
Share of firms introducing new machinery	0.436 (0.076)***	0.387 (0.064)***	0.624 (0.052)***	0.490 (0.074)***
Share of firms aiming to reduce labour cost	−0.297 (0.129)***	−0.125 (0.087)	−0.097 (0.113)	0.077 (0.095)
Total expenditure on innovation per employee	−0.272 (0.104)**	−0.176 (0.081)**	−0.104 (0.133)	0.370 (0.209)*
Constant	14.620 (1.374)***			
Country dummies	Yes			
F test	89.94			
p value	0.0000			
N obs	531			
R2	0.86			

Dependent variable: share of innovative firms. OLS with robust and clustered standard errors and heterogeneous coefficients. Standard errors in parentheses

* Significant at the 90 % level; ** significant at 95 %; *** significant at 99 %

Specialised Suppliers and the Scale and Information Intensive groups. The total expenditure per employee ends up capturing the heterogeneity within industries, resulting in a negative and significant sign for Science Based and Specialised Suppliers.

SB and SS industries show a large role of R&D and the importance of new products, while SII and SD are more oriented towards process innovation through the adoption of new machinery (the coefficients are almost twice the corresponding ones for SS and SB).

In the Appendix Table 11 we report the same regression for services industry only: it can be seen that qualitatively the main results do not change.

We have run the same regression for the five class taxonomy proposed in Tidd et al. (2005) with similar results; for the Information Intensive class the only additional finding is the role played by the aim of opening up new markets, a feature in line with the conceptualization of this type of market services. However, a Wald test does not reject at 1 % level the equality of coefficients between Information Intensive and Scale Intensive (for the Tobit model with bootstrapped standard errors the chi^2 stat is 13.9). Therefore the robustness of the SII group is confirmed.

An additional way to explore the determinants of innovative performance is to use as dependent variable the share of turnover from new products and to test the role of total innovative expenditure, share of firms with product innovation and demand (rate of growth of value added) in each of the four Revised Pavitt classes.

Table 7 The determinants of innovative turnover

	(1) SB	(2) SS	(3) SII	(4) SD
Share of firms introducing new products	0.404 (0.050)***	0.417 (0.039)***	0.179 (0.042)***	0.260 (0.040)***
Total expenditure on innovation per employee	0.180 (0.220)	0.225 (0.189)	0.623 (0.283)**	−0.288 (0.206)
Rate of growth of value added SB	0.168 (0.248)	0.060 (0.379)	0.397 (0.184)**	−0.180 (0.079)**
Constant	5.670 (1.020)***			
N obs	518			
R2	0.41			

Dependent variable: share of turnover from new products. OLS with robust and clustered standard errors and heterogeneous coefficients. Standard errors in parentheses

* Significant at the 90 % level; ** significant at 95 %; *** significant at 99 %

The results for OLS with robust and clustered standard errors are reported in Table 7. As before, we do not report the results of the Tobit because the coefficients are the same.

The Wald test rejects the hypothesis of non-systematic difference among the coefficients for the various classes (χ^2 348.68; p 0.00). For Science Based and Specialised Suppliers industries the main (and only) driver is product innovation. New products are also relevant (with much lower coefficients) for all other groups. In Scale and Information Intensive industries demand growth (reflecting the emergence of new service markets) and total innovation expenditure (reflecting the relevance of new machinery) are also significant. In Suppliers Dominated industries new products are combined with demand reduction, reflecting the pressure to innovate in declining markets. Again, in the Appendix (Table 12), we report the same regression for services industries only, showing that the main results still hold qualitatively.

We have again tested here the robustness of the five class taxonomy, finding that the differences between Information Intensive and Scale Intensive are again rejected (for the Tobit model, the χ^2 is 7.80, p 0.05). We therefore have consistent evidence in favour of the strength of the four Revised Pavitt Classes.

5.2 The relationships with productivity and employment

In Bogliacino and Pianta (2010, 2011) the Revised Pavitt Taxonomy has been applied to the study of the determinants of employment and productivity using the same dataset. In the former work, a model is investigated where total labour input (number of hours worked) is regressed over innovative variables, demand, wages and a variable capturing industrial dynamism. Innovation is broken down into variables reflecting the two competitiveness strategies, discussed in Sect. 4 above. Results show that differences between manufacturing and services are negligible, while a substantial diversity is associated to the Revised Pavitt Classes. In Science

Based industries technological competitiveness is a major factor behind the increase of hours worked. The labour market functioning for this group appears different from the "neoclassical" one, since wage growth is associated to R&D investment, largely spent in research personnel, and no negative correlation with employment growth is found. Specialised Suppliers display a key role of demand (because of the interaction with clients). Scale and Information Intensive are prevalently oriented towards restructuring through labour saving innovation (cost competitiveness). Finally, Suppliers Dominated have a more traditional labour market, a prevalence of process innovation and a strong role of demand.[15]

In the article exploring the determinants of productivity growth (Bogliacino and Pianta 2011), the results show that R&D is the most important driver for Science Based; for SII the drivers are medium level human capital and labour saving strategy; for SS the role of interaction with clients has a positive impact and R&D a non-negligible role; finally, in SD process innovation dominates. In this variety of applications, substantial differences in the types of relationships have emerged; the determinants of innovation performance, productivity and employment growth do differ across Revised Pavitt Classes.

6 Conclusions

In analysing the heterogeneity of innovation and performance across manufacturing and service industries, we have shown that the Revised Pavitt Taxonomy we adopt is a highly effective tool for identifying key differences in: (a) levels and types of innovative efforts; (b) proximity in a multi-dimensional technological domains; (c) determinants of innovative performances; (d) the relationships between innovation and economic performance. The integration we suggest of service industries into the original four class Pavitt Taxonomy for manufacturing has proved to be robust to a great number of statistical tests, using variables drawn from three CIS waves for eight European countries.

The dynamics of technological change in advanced countries appears to be captured in an appropriate way by the distinction among the four Revised Pavitt Classes, highlighting the specificity of innovative processes in each industry group. The insights offered 30 years ago by the Pavitt taxonomy continue to be relevant for understanding today's economies. For Science Based, Specialised Suppliers and Suppliers Dominated classes the feature originally identified still hold; Scale Intensive manufacturing industries have become increasingly similar to Information Intensive services, suggesting that a new, broader class of Scale and Information Intensive sectors can be identified—conceptually and empirically.

Moreover, we have moved beyond the original approach by Pavitt by testing the determinants of innovative performances, showing that different types of technological efforts—starting from the distinction between technological and cost competitiveness—shape the specific patterns of technological change in the four

[15] Tiffin (2014) refers to this work as evidence that competitiveness differs across industry groups and cannot be reduce to price factors.

industry groups. This also applies to the links between the variety of innovative activities, demand factors and economic performance measured in terms of productivity and employment growth.

Our findings have important implications for research and policy. Models of innovation and growth should reflect the diversity of relationships captured by the Revised Pavitt Taxonomy. On the basis of such evidence, the assumption that a single general relationship can effectively describe the behaviour of the whole economy, across manufacturing and service industries, appears to be less and less justified.

On the other hand, Revised Pavitt Classes are able to account for a significant part of existing heterogeneity; in analyses at the industry level, they make a novel connection possible between the processes of technological and structural change.

We claim that this taxonomy may be a valid contribution to the innovation literature, in particular with respect to the identification of key drivers of technical change and in exploring the heterogeneity in the innovation-performance relationship. We think that future research agenda should be addressing the robustness of the taxonomy to new techno-economic paradigms: as an example, one legitimate research question is how to meaningful characterize the eco-innovation, is it homogeneous across Pavitt classes? Secondly we think that both small firms and high growth firms' behavior should be tested across the classes, to assess importance of this kind of technological heterogeneity.

Finally, in terms of policy, one-size-fits-all approaches—that have deeply influenced innovation and industrial policies in advanced countries in the past decades—are clearly undermined by the evidence provided.[16] For innovation, each industry group needs tailor made measures supporting the accumulation of specific knowledge and competences and the strengthening of innovative capabilities, through a combination of supply push, user-producer interaction and demand pull. For upgrading the economic structure, targeted industrial policies that take into account the specific factors supporting growth and employment in each industry group are required. The approach we have developed in this article allows a novel understanding of the complexity of innovation processes in modern economies, while, at the same time, providing a manageable way for addressing heterogeneity, interpreting change, and inspiring policies.

Appendix

See Tables 8, 9, 10, 11 and 12.

[16] The literature on national and sectoral systems of innovation (Malerba 2004) has long pointed out the policy implications of the diversity of innovative dynamics across industries. The importance of active and selective innovation and industrial policy has been argued by Cimoli et al. (2009), Mazzucato (2013) and Pianta (2014).

Table 8 The Revised Pavitt taxonomy for manufacturing and services

Revised Pavitt Taxonomy	NACE Rev. 2
Science based	
Manufacture of chemicals and chemical products	20
Manufacture of basic pharmaceutical products and pharmaceutical prep.	21
Manufacture of computer, electronic and optical products	26
Telecommunications	61
Computer programming, consultancy and related activities	62
Scientific research and development	72
Specialised suppliers	
Manufacture of electrical equipment	27
Manufacture of machinery and equipment n.e.c.	28
Manufacture of other transport equipment	30
Repair and installation of machinery and equipment	33
Real estate activities	68
Legal and accounting activities	69
Management consultancy activities	70
Architectural and engineering activities; technical testing and analysis	71
Advertising and market research	73
Other professional, scientific and technical activities	74
Rental and leasing activities	77
Office administrative, office support and other business support activities	82
Scale and information intensive	
Manufacture of paper and paper products	17
Printing and reproduction of recorded media	18
Manufacture of coke and refined petroleum products	19
Manufacture of rubber and plastic products	22
Manufacture of other non-metallic mineral products	23
Manufacture of basic metals	24
Manufacture of motor vehicles, trailers and semi-trailers	29
Publishing activities	58
Audiovisual activities	59
Broadcasting activities	60
Information service activities	63
Financial service activities, except insurance and pension funding	64
Insurance, reinsurance and pension funding, except compulsory social security	65
Activities auxiliary to financial services and insurance activities	66
Suppliers dominated	
Manufacture of food products	10
Manufacture of beverages	11
Manufacture of tobacco products	12
Manufacture of textiles	13
Manufacture of wearing apparel	14
Manufacture of leather and related products	15
Manufacture of wood and of products of wood and cork, except furniture	16

Table 8 continued

Revised Pavitt Taxonomy	NACE Rev. 2
Manufacture of fabricated metal products, except machinery and equipment	25
Manufacture of furniture	31
Other manufacturing	32
Wholesale and retail trade and repair of motor vehicles and motorcycles	45
Wholesale trade, except of motor vehicles and motorcycles	46
Retail trade, except of motor vehicles and motorcycles	47
Land transport and transport via pipelines	49
Water transport	50
Air transport	51
Warehousing and support activities for transportation	52
Postal and courier activities	53
Accommodation and food service activities	55–56
Veterinary activities	75
Employment activities	78
Travel agency, tour operator reservation service and related activities	79
Security and investigation activities	80
Services to buildings and landscape activities	81

Two digit Nace classification, Rev. 2

Table 9 A multinomial logit analysis for the Revised Pavitt taxonomy: relative risk ratio

	(1) SB	(2) SII	(3) SS
Share of firms introducing new products	0.0838*** (0.0194)	−0.0188 (0.0169)	0.0304 (0.0200)
Share of turnover from new products	0.0810*** (0.0189)	0.00256 (0.0180)	0.0605*** (0.0165)
In-house R&D expenditure (thousand euros per empl)	1.099*** (0.327)	0.773** (0.318)	1.044*** (0.330)
Share of firms introducing new process	−0.0353 (0.0233)	0.0610*** (0.0158)	−0.0618** (0.0241)
Share of firms aiming to reduce labour cost	−0.0685*** (0.0149)	−0.0176 (0.0126)	−0.0151 (0.0128)
Machinery expenditure (thousand euros per empl)	0.139 (0.202)	0.533*** (0.178)	−0.250 (0.213)
Demand: rate of growth of value added	0.131*** (0.0418)	0.0905** (0.0448)	0.113*** (0.0343)
Constant	−4.883*** (0.571)	−2.517*** (0.401)	−1.599*** (0.450)
Wald	169.35***		
Pseudo R2	0.32		
Observations	428		

Dependent variable: Revised Pavitt classes. RRR expressed relatively to the Suppliers Dominated industries. Standard errors in parentheses

* Significant at the 90 % level; ** significant at 95 %; *** significant at 99 %

Table 10 A multinomial logit analysis for the Revised Pavitt taxonomy: odds ratio

	(1) SB	(2) SII	(3) SS
Share of firms introducing new products	0.214*** (0.0544)	0.0768** (0.0391)	0.0949* (0.0528)
Share of turnover from new products	0.176*** (0.0521)	0.0821* (0.0486)	0.0490 (0.0495)
In-house R&D expenditure (thousand euros per empl)	0.621** (0.308)	0.178 (0.306)	−0.956 (0.796)
Share of firms introducing new process	−0.138** (0.0665)	−0.00759 (0.0481)	−0.0756 (0.0539)
Share of firms aiming to reduce labour cost	0.0185 (0.0598)	0.0304 (0.0507)	−0.0175 (0.0595)
Machinery expenditure (thousand euros per empl)	−0.216 (0.443)	0.144 (0.118)	−0.940 (0.633)
Demand: rate of growth of value added	0.0281 (0.0911)	0.00724 (0.0980)	0.00422 (0.0587)
Constant	−6.529*** (1.191)	−3.581*** (0.946)	−0.971 (0.612)
Pseudo R2	0.37		
Wald	75.16***		
Observations	115		

Services industries only. Dependent variable: Revised Pavitt classes. Odds ratio expressed relatively to the Suppliers Dominated industries. Standard errors in parentheses

* Significant at the 90 % level; ** significant at 95 %; *** significant at 99 %

Table 11 The determinants of the share of innovative firms

	(1) SB	(2) SS	(3) SII	(4) SD
Share of firms performing in house R&D	0.254** (0.110)	0.214** (0.106)	0.0782 (0.0712)	0.297*** (0.103)
Share of firms aiming to open new markets	0.766*** (0.224)	0.591*** (0.217)	0.636*** (0.185)	0.118 (0.0979)
Share of firms introducing new machinery	0.623*** (0.0979)	0.587*** (0.111)	0.700*** (0.0779)	0.516*** (0.103)
Share of firms aiming to reduce labour cost	−0.486 (0.325)	−0.0597 (0.326)	0.0550 (0.215)	0.437** (0.176)
Total expenditure on innovation per employee	−0.475** (0.225)	−1.033 (1.115)	−0.144 (0.178)	0.178 (0.352)
Constant	13.77*** (2.840)			
Country dummies	Yes			
F test	102.63			
p value	0.0000			
N obs	153			
R2	0.91			

Services industries only. Dependent variable: share of innovative firms. OLS with robust and clustered standard errors and heterogeneous coefficients. Standard errors in parentheses

* Significant at the 90 % level; ** significant at 95 %; *** significant at 99 %

Table 12 The determinants of innovative turnover

	(1) SB	(2) SS	(3) SII	(4) SD
Share of firms introducing new products	0.241** (0.0952)	−0.0287 (0.150)	0.0966* (0.0564)	0.0884 (0.0571)
Total expenditure on innovation per employee	1.141*** (0.256)	−0.295 (3.396)	0.103 (0.278)	−0.00495 (0.221)
Rate of growth of value added SB	0.439 (0.468)	1.161 (0.795)	0.627*** (0.217)	−0.214 (0.184)
Constant	7.27 (1.57)***			
N obs	130			
R2	0.41			

Services industries only. Dependent variable: Share of turnover from new products. OLS with robust and clustered standard errors and heterogeneous coefficients. %. Standard errors in parentheses

* Significant at the 90 % level; ** significant at 95 %; *** significant at 99 %

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